

MOI GIRLS' ELDORET 2025 PREDICTION



POSSIBLE KCSE EXAM MATHEMATICS

121/1

PAPER 1

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO THE CANDIDATES

- a) Write your **name** and **Random no.** the spaces provided above.
- b) Sign and write **date** of examination in the spaces provided above.
- c) This paper consists of **two** sections; **Section I** and **Section II**.
- d) Answer **All** questions in **Section I** and only **Five** questions from **section II**
- e) Show all the steps in your calculations giving answers at each stage in the spaces provided below each question.
- f) Marks may be given for correct working even if the answer is wrong.
- g) Non-programmable silent electronic calculators and KNEC Mathematical tables may be used except where stated otherwise.

For examiner's use only.

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

GRAND

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TOTAL

1. Evaluate $\frac{-4 \text{ of } (-4 + -15 \div 5 + -3 - 4 \div 2)}{84 \div -7 + 3 - -5}$ (3 mks)

2. Simplify $\frac{9x^2 - 1}{3x^2 + 2x + 1}$ (2 mk)

3. Solve the following inequality and state the integral solutions.

(3 marks)

$$\frac{1}{2}(24 - 4x) > 6\left(3x - \frac{4}{3}\right) \geq -\frac{2}{3}(42 + 3x)$$

4. The position vector of P is $\mathbf{OP} = 2\mathbf{i} - 3\mathbf{j}$ and M is the mid – point of PQ. Given $\mathbf{OM} = \mathbf{i} + 4\mathbf{j}$, Obtain the vector $\mathbf{PQ}$.

(3 marks)

5. Use tables of reciprocals only to work out.

(3 mks)

$$\frac{5}{0.0396} + \frac{12}{0.593}$$

6. A straight line passes through points A (-2,6) and B (4, 2).

(a) M is the midpoint of line AB. Find the coordinate of M.

(2 mks)

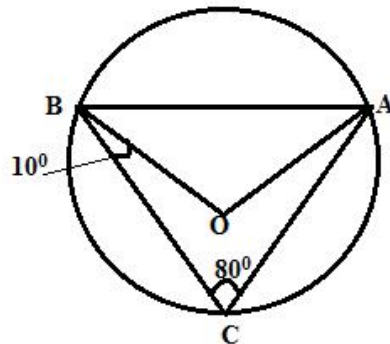
(b) Determine the equation of a straight line passing through point M and is perpendicular to AB.

(2 mks)

7. An open right circular cone has radius of 5cm and a perpendicular height of 12cm. Calculate the surface area of the cone. (take $\pi=3.142$). (3 mks)

8. **Mary** spends a total of sh. 970 on buying 3 text books and 5 pens. If she had bought 2 textbooks and 8 pens, she would have saved sh. 90. Find the cost of one textbook. (3 mks)

9. In the figure below O is the centre of the circle. $\angle BCA = 80^\circ$ and $\angle CBO = 10^\circ$. Determine the size of $\angle CAB$. (3 mks)



10.In a bookstore, books packed in cartons are arranged in rows such that there are 50 cartons in the first row, 48 cartons in the next row, 46cartons in next and so on.

(a) How many cartons will be there in 8th row. **(2 mks)**

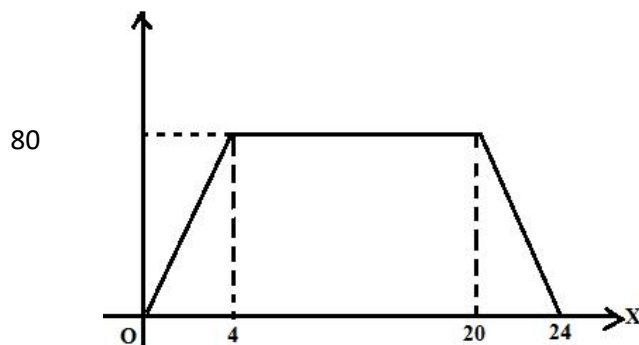
(b) If there are 20 rows in total, find the total number of cartons in the books store. **(2 mks)**

11.Find the value of x if. **(3 mks)**

$$\left(\frac{27}{8}\right)^{x+7} = \left(\frac{4}{9}\right)^{-3x}$$

12.The image of a point K (1,2) after translation is K¹ (-1,2). What is the coordinate of the point R whose image is R¹ (-3,3) after undergoing the same translation? **(3 mks)**

13. The figure below is a velocity time graph for a car.



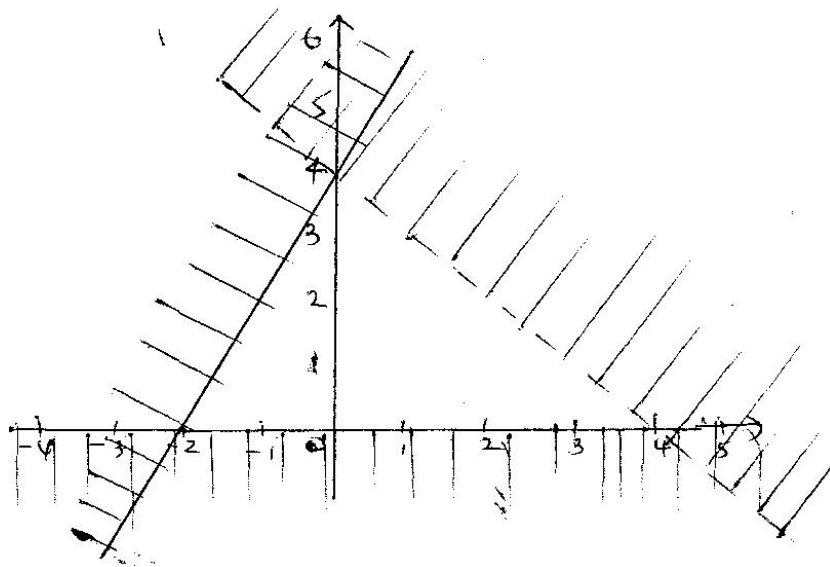
(a) Find the total distance travelled by the car (2 mks)

(b) Calculate the deceleration of the car. (2 mks)

14. Security light poles have been erected along both sides of a street in Bahati town. The poles are 50m apart along the left-hand side of the road while they are 80m apart along the right-hand side. At one end of the road the poles are directly opposite each other. How many poles will be erected by time the poles are directly opposite each other at end of the road? (3 mks)

15. The exterior angle of a regular polygon is equal to one third of the interior angle. Calculate the number of sides of the polygon. (3 mks)

16. Write down the inequalities that define the unshaded region marked R in the figure, below. (3 mks)



SECTION B

17. Nakuru county government is to construct a floor of an open wholesale market whose area is 800m^2 .

The floor is to be covered with a slab of uniform thickness of 200mm. In order to make the slab, sand, cement and ballast are to be mixed such that their masses are in the ratio 3:2:3 respectively. The mass of dry mass of dry slab of volume 1m^3 is 200kg.

(a) Calculate

(i) The volume of the slab. **(2 mks)**

(ii) The mass of the dry slab. **(2 mks)**

(iii) The mass of cement to be used. **(2 mks)**

(b) If one bag of cement is 50kg, find the number of bags to be purchased. **(1 mk)**

- (c) If a lorry carries 10 tonnes of ballast, calculate the number of lorries of ballast to be purchased. (3 mks)

18. Paul is a sales executive earning sh 20,000 and a commission of 8% for the sales in excess of 100,000. In January 2014 he earned a total of 48000 in salaries and commissions.

- (a) Determine the amount of sales he made in that month. (4 mks)

(b) If the total sales in the month of February and march increased by 18% and then dropped by 25% respectively. Calculate.

- (i) Paul's commission in the month of February. (3 mks)

(b) His total earnings in the month of March.

(3 mks)

19.Two tanks are similar in shape. The capacity of the tanks are 1,000,000 litres and 512,00 litres respectively.

(a) Find the height of the smaller tank if the larger one is 300cm tall. **(5 mks)**

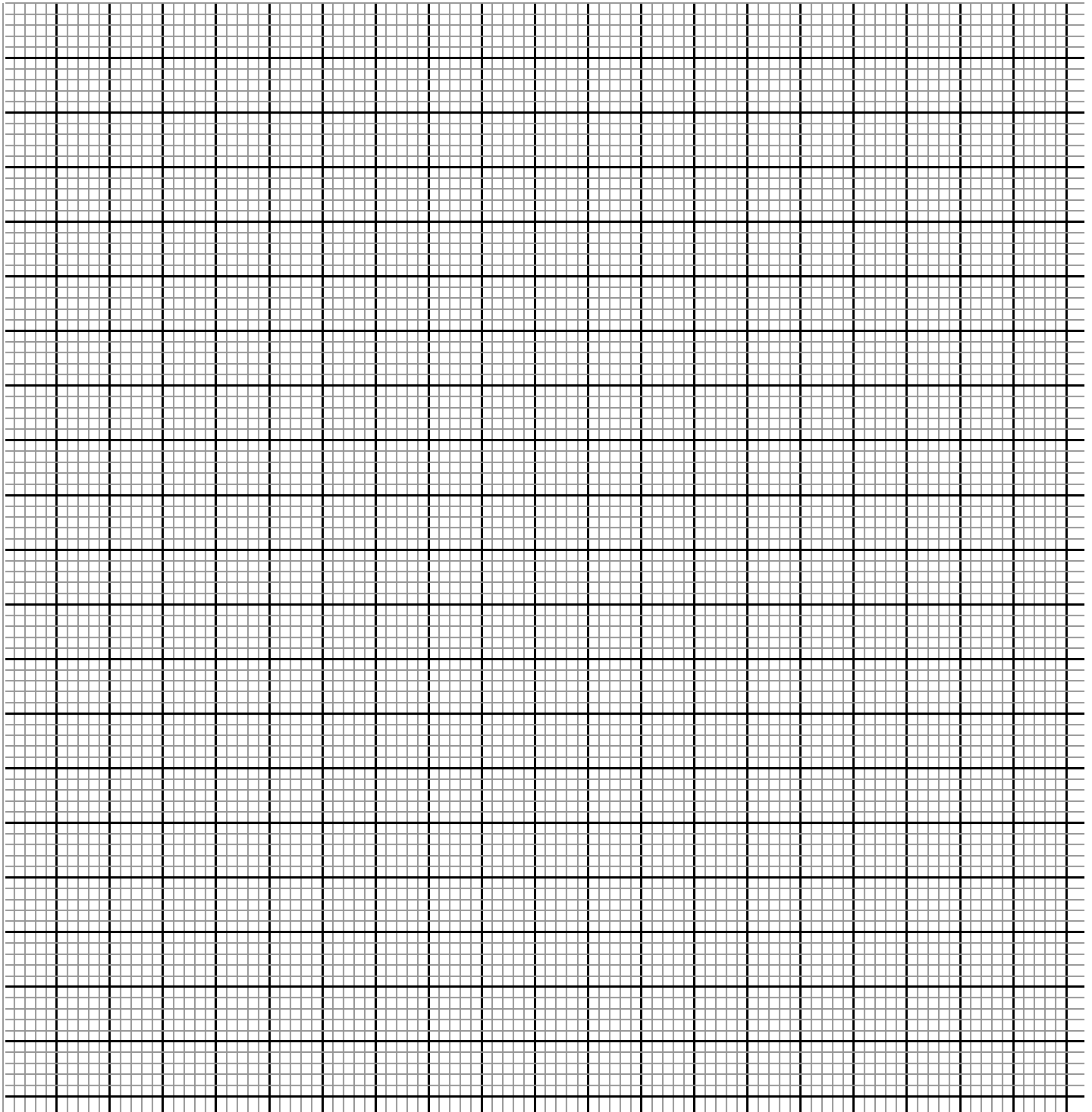
(b) Calculate the surface area of the tank if the smaller one has a surface area of 768cm^3 **(3 mks)**

(c) Calculate the mass of the larger tank if the mass of the larger one is 800kg. **(2 mks)**

20. The vertices of a triangle ABC are A (2,5) B (4,3) and C (2,3). It rotates half-turn about the origin.

(a) Draw triangle ABC and $A^1B^1C^1$ under it.

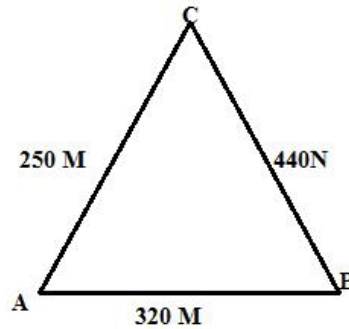
(4 mks)



(b) The image $A'B'C$ is mapped onto $A''B''C''$ under a reflection R in the line $x=0$ followed by a translation $T = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$. Find the coordinates of $A''B''$ and C'' and $A'''B'''C'''$. Hence draw triangle $A'''B'''C'''$. **(4 mks)**

(c) Find the area of the triangle $A''B''C''$. **(2 mks)**

21. Ombati owns a farm that is triangular in shape as shown below.



(a) Calculate the size of angle BAC. **(2 mks)**

(b) Find the area of the farm in hectares. **(3 mks)**

(c) Ombati wishes to irrigate his farm using a sprinkler machine in the farm such that it is equidistant from points A, B and C.

(i) The sprinkler rotates in a circular motion so that the maximum point reached by the water jets is the vertices A, B and C. Calculate the area outside the farm that will be irrigated. **(5 mks)**

22.Trasnsline bus left Nairobi at 8.00 am and travelled to Nakuru at an average speed of 80km/h. A car left Nakuru at 3.30 am and travelled to Nairobi at an average speed of 120km/h. Given that the distance between Nairobi and Nakuru is 400km, Calculate.

(a) The time the car arrived in Nairobi. **(3 mks)**

(b) The time the two vehicles met. **(4 mks)**

(c) The distance from Nairobi to the meeting point. **(2 mks)**

(d) The distance of the bus from Nakuru when the car arrived in Nairobi.
(2 mks)

23. Town B is 102km on the bearing of 122° from town A. Town C is 94 km on bearing of 062° from B. Town D is on a bearing of 073° from A and 336° from C.

- (a)** Using a scale of 1cm to represent 20km, draw a scale diagram to show the relative positions of town A, B, C and D. **(4 mks)**

- (b)** Using your diagram, determine.

(i) The bearing B from D. **(1 mk)**

(ii) The bearing of A from C. **(1 mk)**

(iii) The distance from town A to D. **(1 mk)**

(iv) The distance from town B and D.

(1 mk)

24. The table below gives some of the values of x and y for the functions $y = \frac{1}{2}x^2 + 22x + 1$ in the interval $0 \leq x \leq 6$.

(a)

x	0	1	2	3	4	5	6
y	1					23.5	

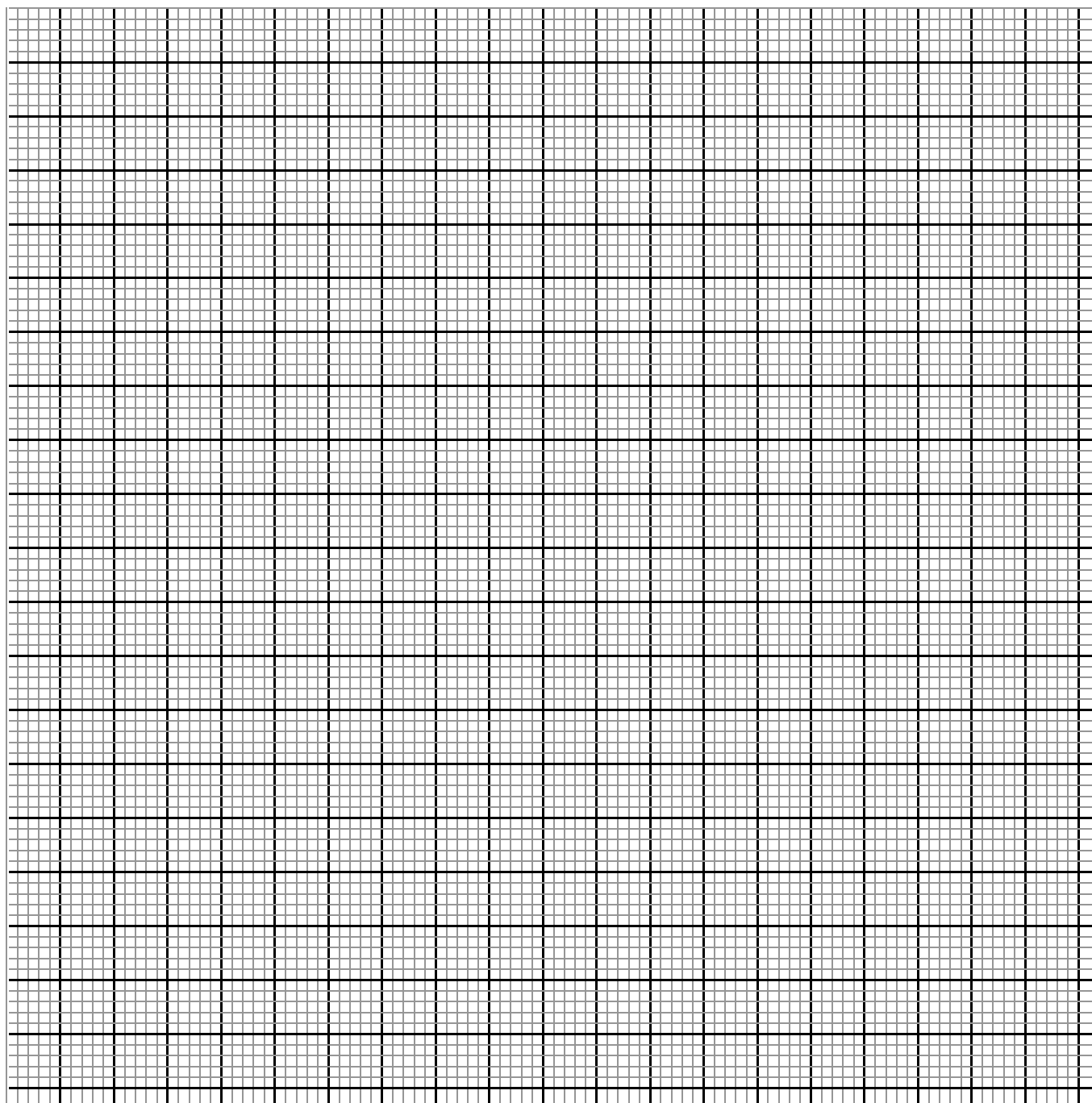
Complete the values in the table above.

(1 mk)

(b) Use the values in the

table to draw the graph of function on the grid provided below.

(2 mks)



- (b) Using the graph and the mid-ordinate rule with 6 stripes, estimate the area bounded by the curve, the x-axis, the y-axis and the line $x=6$. (3 mks)

- (d) Using integration, calculate the exact area and hence find percentage error made when mid-ordinate rule is used. Give your answer correct to 2.dp. **(4 mks)**